

ROHS SOLDER TYPES

Comprehensive Guide for PCBA Manufacturing

Lead-Free Solder Alloys | SAC305 | SnCu | SnAgCu | Process Guidelines

Prepared for PCBA Assembly & Quality Standards Compliance
Reference: IPC/J-STD-001 & RoHS Directive 2011/65/EU

1. INTRODUCTION TO ROHS SOLDER

The Restriction of Hazardous Substances (RoHS) Directive mandates the elimination of lead (Pb) and other hazardous materials from electrical and electronic equipment. In PCBA manufacturing, this has driven the transition from traditional Sn63Pb37 (eutectic tin-lead) solder to lead-free alternatives. This guide covers the major RoHS-compliant solder types used in modern electronics assembly.

Key RoHS Solder Requirements:

- Lead content must be < 0.1% (1000 ppm) by weight
- Higher melting points require adjusted thermal profiles
- Compatible flux systems for improved wetting
- Component and PCB materials must withstand elevated temperatures

Note: Lead-free solders typically melt at **217–227°C**, compared to 183°C for Sn63Pb37. This 30–40°C increase requires careful process optimization to prevent component damage while ensuring proper wetting.

2. MAJOR ROHS-COMPLIANT SOLDER ALLOYS

2.1 SAC305 (Sn96.5/Ag3.0/Cu0.5)

SAC305 is the most widely adopted lead-free solder alloy in the electronics industry. It offers an excellent balance of mechanical properties, wetting characteristics, and reliability. The silver content provides strength, while copper improves wetting and reduces dissolution of copper pads.

Properties:

- **Melting Point:** 217–220°C (eutectic near 217°C)
- **Typical Reflow Peak:** 245–250°C
- **Wave Solder Pot:** 250–260°C
- **Wetting Speed:** Medium (slower than SnPb, faster than SnCu)
- **Joint Appearance:** Dull/grainy (normal for lead-free)
- **Relative Cost:** 3× compared to Sn63Pb37

Applications:

- SMT reflow soldering (solder paste)
- High-reliability consumer electronics
- Automotive electronics (Tesla, Zoox compliance)
- BGA and CSP area array packages

■■ **Process Note:** SAC305 exhibits higher shrinkage effects and can produce "hot tears" in wave soldering. It also has higher copper dissolution rates (2.1× vs SnPb), requiring more frequent pot analysis and maintenance.

2.2 SnCu (Tin-Copper) & SnCu+Ni (K100)

Sn99Cu1 and variants like **SnCu+Ni (K100)** are cost-effective lead-free alternatives. The addition of Nickel (0.05–0.1%) significantly improves fluidity, wetting, and reduces dross formation. K100 is specifically formulated for wave soldering and hand soldering operations.

Properties:

- **Melting Point:** 227°C (eutectic, sharp melting range)
- **Typical Wave Pot:** 255–265°C
- **Wetting Speed:** Slower than SAC305 (requires longer contact time)
- **Joint Appearance:** Shiny, reflective (better cosmetics than SAC)
- **Copper Dissolution:** Low (1.0× vs SnPb, better than SAC305)
- **Relative Cost:** 1.5× compared to Sn63Pb37
- **Dross Formation:** Lower than SAC305 with proper maintenance

Applications:

- Wave soldering (bar solder)
- Through-hole assembly (PTH)
- Hand soldering and rework operations
- Cost-sensitive high-volume production

■ **Advantage:** SnCu+Ni offers easier pot management, lower reactivity to equipment, and excellent hole-fill for through-hole components when process parameters are optimized. The shiny joint appearance simplifies visual inspection.

2.3 Other RoHS Solder Alloys

SAC387 / SAC405 (Higher Silver)

Higher silver content (3.8–4.0% Ag) improves mechanical strength and thermal fatigue resistance but increases cost and can reduce drop-shock performance. SAC387 was an early candidate but largely superseded by SAC305 for cost reasons.

SAC105 (Low Silver)

With only 1.0% Ag, SAC105 offers **superior drop-shock resistance** compared to SAC305, making it suitable for portable devices (cell phones). However, thermal cycling performance is reduced. Often used with dopants like Manganese (SACM™) to balance properties.

SnBi (Tin-Bismuth) & SnBiAg

Low melting point (~138–213°C) alloys for temperature-sensitive components. ■■ **Caution:** Bismuth is incompatible with lead contamination—mixing forms a low-melting SnPbBi phase at 96°C, causing catastrophic joint failure.

SnZn (Tin-Zinc)

Low-cost alternative with good mechanical properties but poor oxidation resistance. Requires nitrogen reflow atmosphere and specialized flux systems. Limited adoption due to process sensitivity.

3. SOLDER FORMS IN PCBA MANUFACTURING

3.1 Solder Paste (Reflow SMT)

Solder paste consists of **lead-free solder powder** mixed with flux medium. It is stencil-printed onto PCB pads, components are placed, and the assembly passes through a reflow oven. SAC305 Type III or IV powder is standard for most applications.

Key Parameters:

- **Alloy:** SAC305 (96.5% Sn, 3.0% Ag, 0.5% Cu)
- **Powder Size:** Type III (25–45µm) for standard SMT; Type IV (20–38µm) for fine pitch
- **Flux Type:** No-clean ORH0/ORH1, Water-soluble, or Low-residue
- **Storage:** 0–10°C refrigerated; 4–8 hour stencil life after print

3.2 Solder Bar (Wave Soldering)

Solid bar solder is loaded into the wave solder machine pot and melted. The PCB passes over the molten wave to solder through-hole components. SnCu+Ni (K100) bars are preferred for wave due to lower copper dissolution and better cosmetics.

3.3 Solder Wire (Hand Soldering / Rework)

Flux-cored wire solder for manual operations. Rosin-core (R-type) or no-clean flux core is standard for electronics. Acid-core is strictly for plumbing/non-electrical use. SAC305 or SnCu+Ni wire with diameters 0.5–1.2mm are common for PCBA rework.

3.4 Solder Preforms

Precision-shaped solder pieces (rings, washers, rectangles) used for specific applications like connector soldering, thermal interface, or die-attach. Custom alloys including high-temp BiAgX for component package die-attach.

4. ALLOY COMPARISON TABLE

Alloy	Composition	Melt Point	Cost Factor	Best Use
Sn63Pb37	63% Sn, 37% Pb	183°C	1.0× (Baseline)	Legacy / Exempt
SAC305	96.5% Sn, 3.0% Ag, 0.5% Cu	217°C	3.0×	SMT Reflow, High Reliability
SAC387	95.5% Sn, 3.8% Ag, 0.7% Cu	217°C	3.5×	Early RoHS / Japan
SAC105	98.5% Sn, 1.0% Ag, 0.5% Cu	217°C	2.2×	Drop-shock (mobile)
SnCu (K100)	99.4% Sn, 0.6% Cu + Ni	227°C	1.5×	Wave / Hand Solder
Sn99Cu1	99% Sn, 1% Cu	227°C	1.3×	Low-cost wave
SnBi58	42% Sn, 58% Bi	138°C	1.8×	Low-temp / Temp-sensitive
SnBiAg	~91% Sn, 4.8% Bi, 3.4% Ag	213°C	2.0×	Low-temp alternative

Selection Criteria: Choose SAC305 for reflow SMT when reliability is critical. Choose SnCu+Ni (K100) for wave soldering to reduce operating costs and improve cosmetics. Avoid SnBi if any lead contamination risk exists in the process.

5. PROCESS GUIDELINES FOR LEAD-FREE SOLDERING

5.1 Reflow Soldering Profile (SAC305 Paste)

Lead-free reflow requires a higher peak temperature and longer time above liquidus (TAL). Typical profile for SAC305:

Zone	Temperature	Time / Rate
Preheat	150–180°C	60–120 sec
Soak	180–200°C	60–90 sec
Reflow Peak	245–250°C	20–40 sec above 217°C
Cooling	< 4°C/sec	Down to 100°C

5.2 Wave Soldering Parameters

For SnCu+Ni (K100) or SAC305 bar solder in wave machines:

- **Pot Temperature:** 255–265°C (SnCu+Ni) or 250–260°C (SAC305)
- **Contact Time:** 3–5 seconds (longer than SnPb due to slower wetting)
- **Conveyor Angle:** 5–7° for proper drainage
- **Flux:** Alcohol-based water-washable ORH1 or no-clean VOC-free
- **Preheat:** 100–130°C top side before wave

5.3 Hand Soldering / Rework

Lead-free hand soldering requires higher wattage irons (60–80W) and lead-free capable tips. Dwell times are longer (5–7 seconds vs 2–5 seconds for SnPb).

- **Tip Temperature:** 360–400°C (750°F typical)
- **Tip Type:** Lead-free plated (iron/chromium coating)
- **Wire:** SAC305 or SnCu+Ni with rosin/no-clean flux core
- **Technique:** Apply iron to pad/lead, feed solder from opposite side
- **Do NOT retouch** dull lead-free joints—this weakens the intermetallic layer

Tip Life: Lead-free soldering accelerates tip oxidation and de-wetting. Always tin the tip after use, use lower temperatures when possible, and power off when idle.

6. QUALITY & INSPECTION CRITERIA

6.1 Visual Inspection (IPC-610 Class 2/3)

Lead-free joints have inherently **duller, grainier** appearance than SnPb. This is normal and NOT a defect. Acceptance criteria per IPC-A-610:

- **Acceptable:** Matte finish, slight graininess, good wetting angle ($< 90^\circ$)
- **Reject:** Excessive voiding, cracks, cold joints, insufficient wetting
- **Reject:** Hot tears / shrinkage cracks in wave soldered joints

6.2 Common Lead-Free Defects

- **Tombstoning:** SAC305 with high Ag can cause passive components to stand upright
- **Hot Tears / Shrinkage:** SAC305 in wave soldering; switch to SnCu+Ni to reduce
- **Non-wetting:** Due to oxidation or insufficient flux activation—check thermal profile
- **Copper Dissolution:** Excessive Cu pad erosion in SAC305 wave pots—monitor Cu levels
- **Pad Lifting:** Higher temperatures can delaminate pads on poor-quality PCBs

6.3 X-Ray & Cross-Section Analysis

For BGA and hidden joints, X-ray inspection checks voiding ($< 25\%$ area acceptable per IPC). Cross-sectioning verifies intermetallic compound (IMC) layer thickness—target $1\text{--}3\mu\text{m}$. Excessive IMC growth ($> 5\mu\text{m}$) indicates overheating and reduces reliability.

7. HANDLING, STORAGE & CONTAMINATION CONTROL

7.1 Solder Paste Storage

- **Refrigeration:** 0–10°C sealed container; do NOT freeze
- **Shelf Life:** 6 months refrigerated; 4–8 hours at room temp after opening
- **Thawing:** 4 hours at room temperature before use; do NOT force-thaw
- **Stirring:** Hand-stir or machine mix before stencil print

7.2 Bar Solder Pot Management

- **Lead Contamination:** Keep leaded and lead-free lines physically separated
- **Copper Analysis:** Check Cu levels weekly; SAC305 dissolves Cu 2× faster than SnPb
- **Dross Removal:** Skim dross daily; SnCu produces ~2× dross vs SnPb
- **Pot Balancing:** Add pure Sn or alloy bars to maintain composition after analysis

7.3 Wire Solder Handling

- Store in dry environment; flux core can dry out in humidity
- Keep lead-free wire clearly labeled and separated from SnPb wire
- Use within 2 years of manufacture for optimal flux activity

■■ **Critical:** Even trace lead contamination (> 1000 ppm) in a lead-free pot voids RoHS compliance. Use dedicated tools, pots, and irons for lead-free operations. Implement color-coding (e.g., GREEN for lead-free).

8. QUICK REFERENCE SUMMARY

For SMT Reflow:

- Use **SAC305 paste** (Type III/IV powder) with no-clean or water-washable flux
- Peak reflow 245–250°C, TAL 30–60 seconds

For Wave Soldering:

- Use **SnCu+Ni (K100) bar** for better cosmetics and lower cost
- Pot temp 255–265°C, contact time 3–5 seconds

For Hand Soldering / Rework:

- Use **SAC305 or SnCu+Ni wire** with rosin/no-clean flux core
- Tip temp 360–400°C, dwell 5–7 seconds

For Low-Temperature:

- Use **SnBiAg** only in guaranteed lead-free environments
- Verify no lead contamination risk in process

References:

- IPC-A-610: Acceptability of Electronic Assemblies
- J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
- RoHS Directive 2011/65/EU (Restriction of Hazardous Substances)
- Indium Corporation Technical Papers on SAC Alloys
- Kester Solder Technical Documentation